

Submission PHYS1-PS1-SUB01

Question PHYS1-PS1-Q01

scratch: show step

1. $v = v_{\text{initial}} + at = 3 + 2(5) = 13 \text{ m/s}$

2. $\Delta x = v_{\text{initial}}t + \frac{1}{2}at^2 = 3(5) + \frac{1}{2}(2)(5)^2 = 40 \text{ m}$

=> FINAL: $v = 13 \text{ m/s}$; displacement = 40 m (check units/interpretation)

Question PHYS1-PS1-Q02

scratch: show step

1. $N = mg = 29.4 \text{ N}$; friction = $\mu N = 5.9 \text{ N}$

2. $\Sigma F = 18 - 5.9 = 12.1 \text{ N}$

3. $a = \Sigma F / m = 4.04 \text{ m/s}^2$

=> FINAL: $a = 4.04 \text{ m/s}^2$; friction opposes motion (check units/interpretation)

Question PHYS1-PS1-Q03

scratch: show step

1. $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2 \cdot 9.8 \cdot 7}$

2. $v = 11.71 \text{ m/s}$; mass cancels

=> FINAL: $v = 11.71 \text{ m/s}$ (check units/interpretation)

Question PHYS1-PS1-Q04

scratch: show step

1. $p_{\text{total, before}} = p_{\text{total, after}}$

2. It applies when external impulse on the system is negligible.

=> FINAL: total momentum before equals after when external impulse is negligible (check units/interpretation)

Question PHYS1-PS1-Q05

scratch: show step

1. Impulse is area under $F(t)$: $J = F\Delta t = 6(0.5) = 3 \text{ N}\cdot\text{s}$.

2. $\Delta p = J$, so momentum changes by $3 \text{ kg}\cdot\text{m/s}$ in the force direction.

=> FINAL: impulse = $3 \text{ N}\cdot\text{s}$; momentum changes by $3 \text{ kg}\cdot\text{m/s}$ (check units/interpretation)

Question PHYS1-PS1-Q06

scratch: show step

1. $v = v_{\text{initial}} + at = 4 + 4(3) = 16 \text{ m/s}$

2. $\Delta x = v_{\text{initial}}t + \frac{1}{2}at^2 = 4(3) + \frac{1}{2}(4)(3)^2 = 30 \text{ m}$

=> FINAL: $v = 16 \text{ m/s}$; displacement = 30 m (check units/interpretation)